

Service Under Pressure: Statistical Insights into Match Predictions in Tennis

Chelsea Seegars

Abstract

Service game are essential in determining the outcome of all tennis matches but especially for professional tennis. Often service games show the mental and technical capabilities of players since it's a chance for a player to dictate the start of each point. This study examines three predictors: double faults, break points faced, and aces were used to assess their significance in predicting match outcomes for top 50 ATP and WTA players. Logistic regression models were developed for both the ATP and WTA and other metrics were applied like ROC curves, confusion matrices, and statistical tests were used to evaluate the models' performance and predictive capabilities. Results highlighted gendered differences in the predictor's significance. The findings suggests that break points faces are the most significant predictor across both genders, while aces and double faults have varying levels on influence on match outcome. The results of this study present actionable insights for players and coaches to optimize performance. This research emphasizes the importance of the service game in shaping match outcomes and presents a strong foundation for future studies in predictive tennis analytics.

Introduction

Critical high-pressure moments determine the outcome of a tennis match in professional tennis. Often, the finest margins determine the winner. The Association of Tennis Professionals, frequently shorthand to the ATP, is the governing body for male professional tennis players, and the Women's Tennis Association, often shorthand to the WTA, is the governing body for female professional tennis players. In tennis, a match is composed of three or five sets; within a set, there are at least six games, and a match is won when a player wins most of the sets. In a five-set match, it is best of three, and in a three-set match, it is best of two.

In 2024, even the best players in the ATP and the WTA, like Carlos Alcaraz, Iga Swiatek, Jannik Sinner, and Coco Gauff, did not have a total points won percentage above 57% (Sackmann, 2024). This statistic shows that even the best players barely win most points in a match. Beyond the top 20 players, the total points won percentage drops even lower to around 50%, further proving it is the key moments determining match winners in tennis (Sackmann, 2024). Various metrics influence the outcome of a match, but a player has the most control over a point during their service game (when the player gets to serve). There are multiple components of a service game, but the three main predictors I used are double faults, break points faced, and aces. Double faults are when the serving player makes two consecutive serving errors, and the opposing player is awarded a point. A break point is when the opposing player can win the game by winning the next point. Facing a break point is used when the server is under pressure from the opposing player. Aces is when the serving player serves the ball, and the opposing player does not return the ball. Identifying critical contributors to a match outcome can help players, coaches, and analysts. Highlighting the technical areas that are most critical to match success could improve player consistency. Investigating these factors can help bridge the gap between mental pressures and physical

performance, contributing to more accurate predictive models. This paper aims to determine whether break points faced, double faults or aces are more significant predictors of a loss. If there is a gendered difference similar to Lackner's and Weichselbaumer's paper, where they found that women were more affected by a poor service game than men. There could also be a collinearity problem since break points faced and double faults often represent a poor service game, and poor service games often mean the player will lose.

Statement of Purpose

The purpose of this study is to analyze the role of service-game metrics—double faults, break points faced, and aces—in predicting tennis match outcomes for male (ATP) and female (WTA) professional players. By applying logistic regression and evaluating key performance metrics such as ROC curves, confusion matrices, and model coefficients, this study aims to identify the most significant predictors of match outcomes and explore gendered differences in how these predictors influence performance. Insights from this analysis are intended to inform players, coaches, and analysts on strategies to optimize match performance and better understand pressure-related dynamics in elite tennis.

Literature Review

Psychological Pressure and Compounded Errors During Elite-level Tennis (2021) focuses on how pressure and previous form can impact player performance in Grand Slams. They focused on likelihood of unforced errors and its relationship with pressure. The dataset had over 650,000 points from all the Grand Slams from 2016 to 2019 (Harris et al., 2021). The variables used were which player was serving, which player won a point and the occurrence of clean winners, break points, unforced errors and double faults (Harris et al., 2021). They then analyzed pressure by having a range from 1 to 5. Every point starts with a base pressure of 1 and would increase depending on if it was late in game, if the opponent could take the set, if it's the final deciding set (3rd for womens, 5th for men), if facing a break

point or if the opponent had game point (Harris et al., 2021). After using linear mixed effects models and odds-ratio to examine the dataset (Harris et al., 2021). They found that unforced errors became more common when the pressure level was around 3-5 (Harris et al., 2021). Additionally, a player is more likely to keep having unforced errors after having the first one (Harris et al., 2021). This paper suggests that pressure can still impact even the best athletes (Harris et al., 2021).

Can Barely Winning Lead to Losing? Gender and Past Performance (2023) focuses on how minor negative setbacks within a match can affect player performance. Specifically, how a player reacts after a contested service game. They wanted to see if men and women react differently to have a close win for a game. The authors used double faults as an instrumental variable (Lackner & Weichselbaumer, 2023). Double faults were used because they increase the likelihood of a contested long point game (Lackner & Weichselbaumer, 2023). They used logistic regression where the dependent variable was whether the game led to a break and multiple predictors like the number of points in the game and player ranking (Lackner & Weichselbaumer, 2023). They analyzed over 19,000 tennis matches from 2011 to 2015 and 2017 (Lackner & Weichselbaumer, 2023). They discovered that women are more negatively affected by barely winning than men (Lackner & Weichselbaumer, 2023). Men appeared to be unaffected by having a poor service game (Lackner & Weichselbaumer, 2023). They admit that double faults may be partly endogenous and may not be the best variable to depend on (Lackner & Weichselbaumer, 2023).

Differences in Competition Statistics between Winners and Losers in Male and Female Tennis Players in Olympic Games (2019) focuses on analysing performances to understand what the key factors are between winning and losing a tennis match and whether there is a gendered difference (Fernández-García et al., 2019). They specifically looked at the singles matches during the 2012 London Olympics (Fernández-García et al., 2019). They

analysed serve pace, service game, return game, break points, unforced errors, aces, double faults, net play and serve (Fernández-García et al., 2019). They performed a Mann Whitney U test to analyse the data (Fernández-García et al., 2019). The key factors for men were break points won, first serve points won and return points won on first serve (Fernández-García et al., 2019). The key factor for women was break points won (Fernández-García et al., 2019). This study only looked at grass court performance and other studies on different surfaces found different key factors. They found that for men service game was more important to predicting a win but for women it was their return game that was more important (Fernández-García et al., 2019). This study suggests that there are different styles of play in tennis based on gender (Fernández-García et al., 2019).

Methodology

Tennisabstract.com is a popular data collection site created and maintained by a well-known figure in tennis analytics named Jeff Sackmann (Sloan Sports Analytics Conference, n.d.). He made the datasets publicly available on GitHub, which I later cleaned using the tidyr package in RStudio. The most recent complete season of professional singles tennis was the 2023 season. The population of interest in this study includes professional tennis players competing in the top-tier ATP and WTA tournaments. The target group reflects elite-level tennis athletes whose performance metrics are crucial in understanding match outcomes. There are over 2000 players on the ATP tour and over 1300 on the WTA tour, all ranked based on the WTA's or ATP's ranking system (ATP, 2024; WTA, 2024). The data was filtered to the top 50 players to focus on high-level elite tennis. This reduces the sample size but ensures greater consistency in player quality since players outside the top 50 rarely make deep runs in top-level events. Additionally, matches where there were 0 break points faced, 0 aces, and 0 double faults were omitted. Matches with zero break points faced, aces, or double faults were omitted because they do not contribute meaningful information on how these

factors influence match outcomes. The final dataset had 449 matches for the ATP and 400 matches for the WTA, which creates a total of 1,698 single observations since there are two players in every match. There was no missing data, and I verified ten random observations to ensure the data was still correct after cleaning. The target population is players in the ATP and WTA regardless of ranking competing in the top tournaments (250 events, 500 events, 1000 events and the Grand Slams).

There are three predictor variables being used: one is the number of double faults counted per match (x_1), the number of break points faced per match (x_2) and aces (x_3). All three predictors are quantitative metrics. The response variable is the match's outcome (win or a loss), which is categorical. Since the response is categorical, logistic regression was the logical way to interpret our data. The final regression equation looks like $\log \log \left(\frac{P(win)}{(1-P(win))} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$ where x_1 represents double faults, x_2 represents break points faced and x_3 represents aces. Backward and forward elimination were used to identify the best predictor by iteratively removing and adding variables based on their significance. The packages ggplot2 and lattice were used to create graphs and visualize the data. The packages caret, car, and dplyr were used to test the datasets for further interpretive statistics.

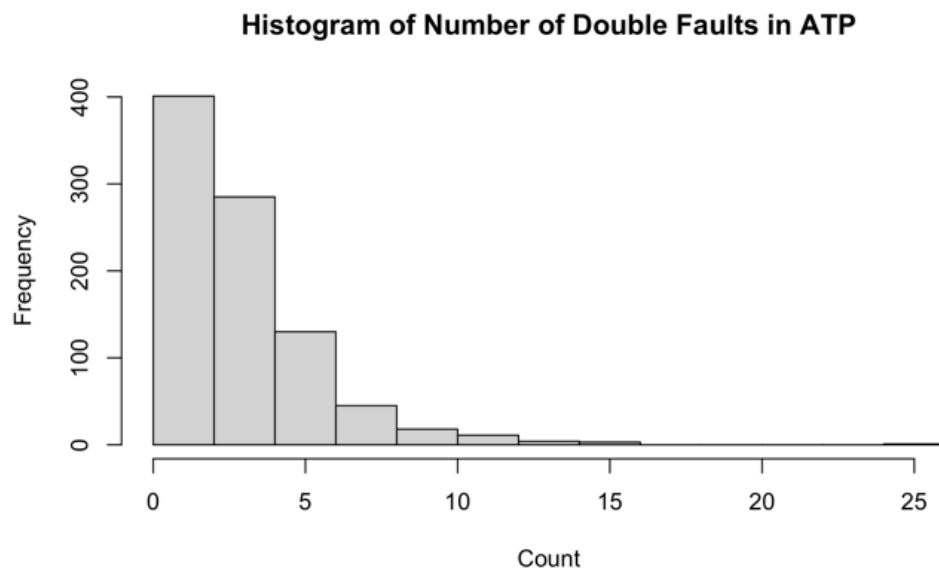
Results

The response variable is the match outcome (win or loss) which is analyzed by three primary predictor variables double faults (x_1), break points faced (x_2) and aces (x_3). There are an equal number of winners and losers and there is no skew for either outcome. These variables were chosen since they reflect key components of a player's service game. During a player's service game, they have control to dictate each point.

The ATP median for double faults is 3 per match, with a range from 1 to 25 (Figure One) and for the WTA the median was 4 with a range from 1 to 17 (Figure Two). The medians are very close with the range for the ATP being wider. The ATP median for number of break points faced is 6 per match with a range from 1 to 26 (Figure Three) for the WTA the median was 8 with a range of 1 to 29 (Figure Four). For double faults the WTA dataset appears slightly more normal than the ATP dataset suggesting that female tennis players are slightly prone to commit double fault errors then their male counterparts. The median for the WTA was also slightly larger also indicating that double faults are more prevalent in the WTA then the ATP. The ATP median for aces is 6 with a range from 1 to 39 (Figure Five) and for the WTA the median was 4 with a range from 1 to 20 (Figure Six). The ATP has a much wider range for aces per match and a larger median implying that aces may be more common in men's tennis then women's.

Figure One

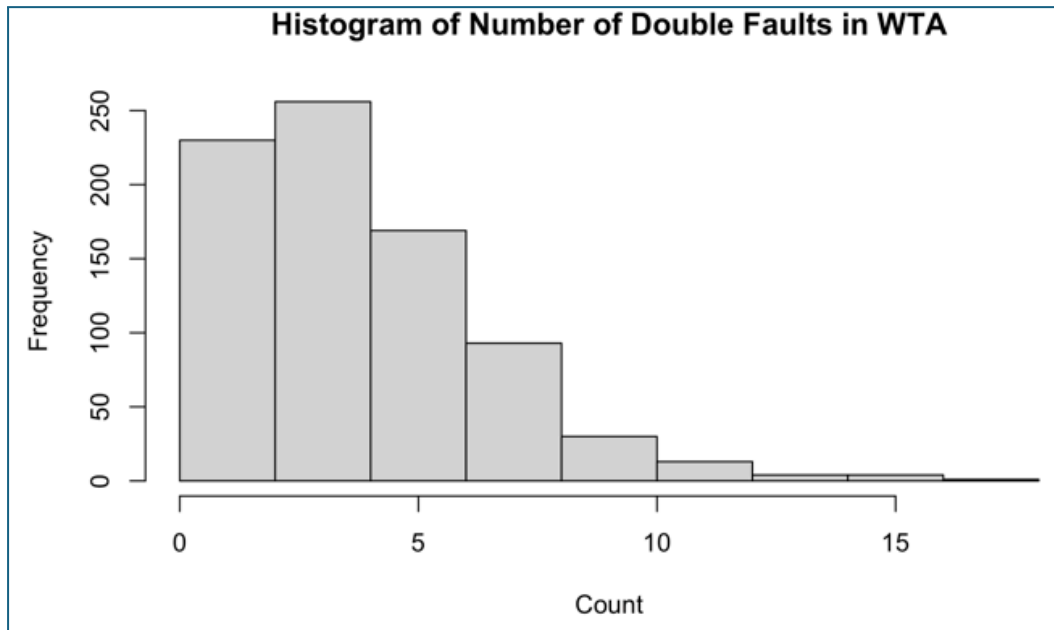
Distribution of Double Faults for ATP Players



Note. Histogram illustrating the distribution of double faults per match for ATP players, with a median of 3 and a range from 1 to 25.

Figure Two

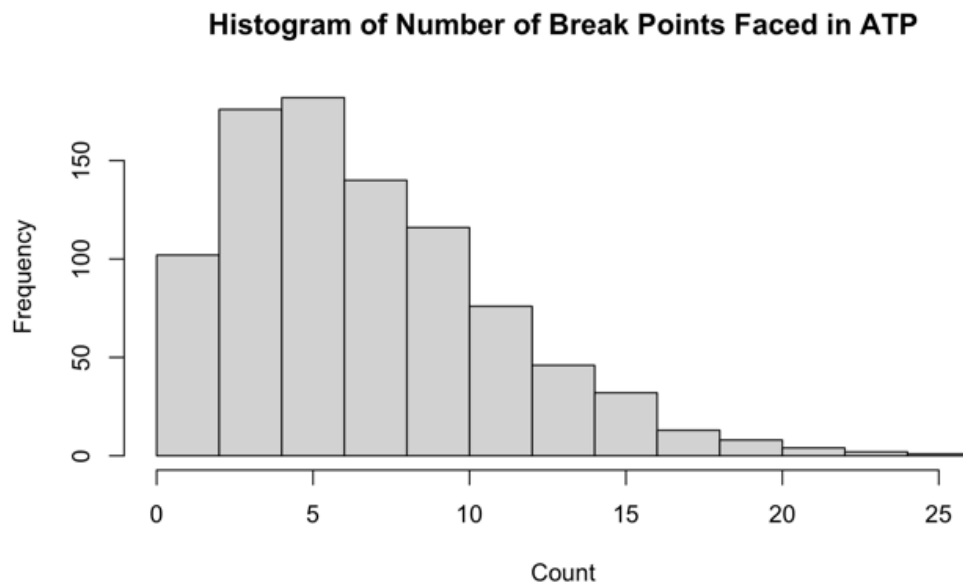
Distribution of Double Faults for WTA Players



Note. Histogram showing the distribution of double faults per match for WTA players, with a median of 4 and a range from 1 to 17.

Figure Three

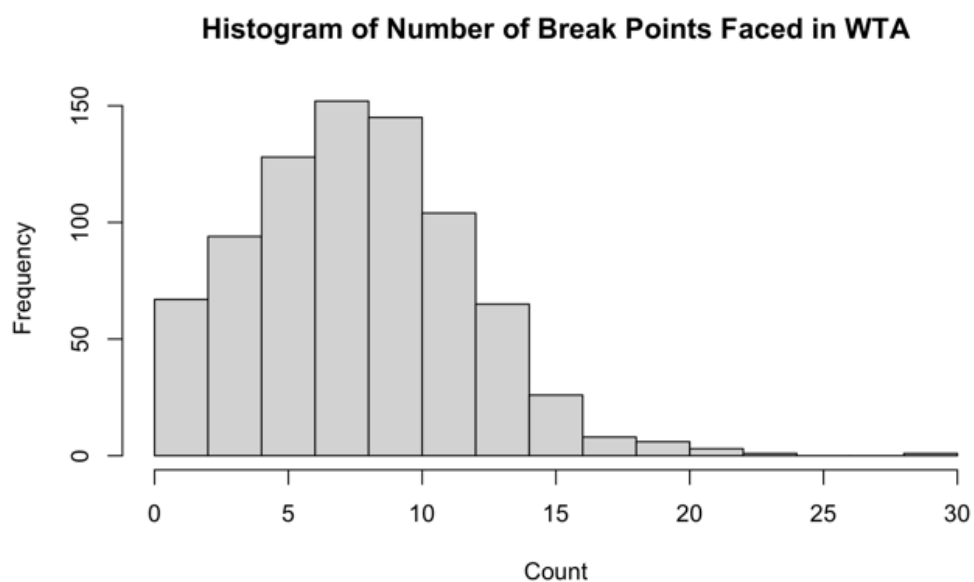
Distribution of Break Points Faced for ATP Players



Note. Histogram depicting the distribution of break points faced per match for ATP players, with a median of 6 and a range from 1 to 26.

Figure Four

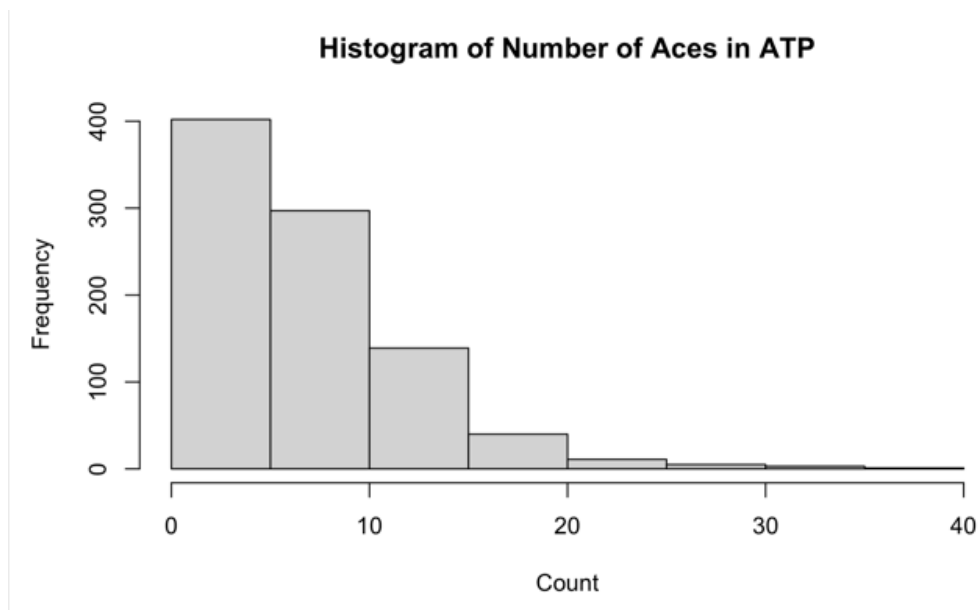
Distribution of Break Points Faced for WTA Players



Note. Histogram of break points faced per match for WTA players, with a median of 8 and a range from 1 to 29.

Figure Five

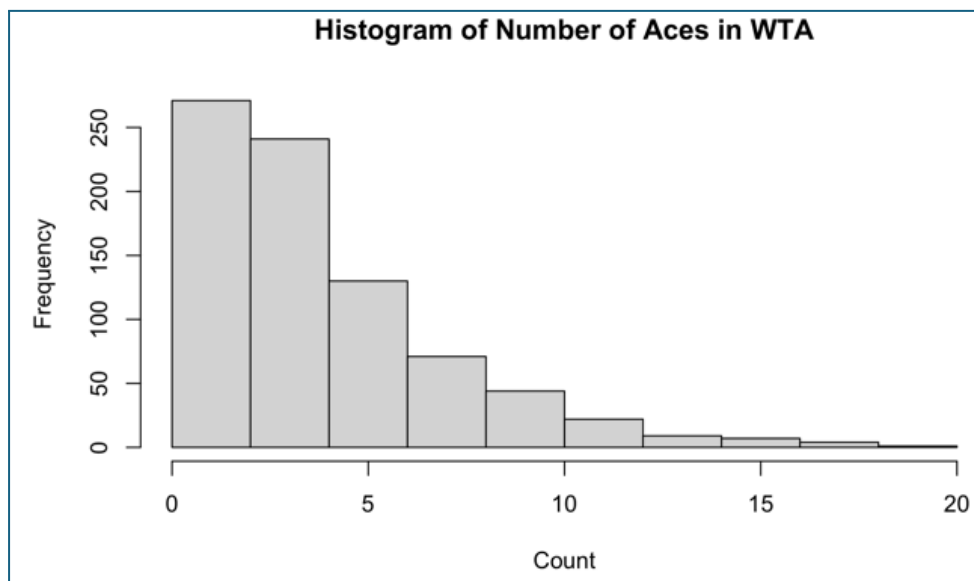
Distribution of Aces for ATP Players



Note. Histogram presenting the distribution of aces per match for ATP players, with a median of 6 and a range from 1 to 39.

Figure Six

Distribution of Aces for WTA Players



Note. Histogram representing the distribution of aces per match for WTA players, with a median of 4 and a range from 1 to 20.

Last season, men hit almost twice the number of aces (7.256) than women (4.351) based off their means (Table One & Two). The ATP and WTA tour did share a similar number of total break points faced and double faults. All three predictors have right skew to their histograms in both the WTA and ATP indicating that most players only commit a few aces, double faults or break points. There is extreme skew in every histogram. Double faults in the ATP had the most extreme skew it was twice as large the skew for the double faults in the WTA, this suggests that WTA players on average commit few double faults per match but that there are some extreme outliers. There was also a large dispersion between the aces in the ATP which is shown by the large standard deviation of aces (5.242) implying that there are some players who score a high number and a low number of aces in a match.

The final regression model is

$\log \log \left(\frac{P(\text{win})}{(1-P(\text{win}))} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$ and the correlations between double faults, break points faced and aces were all weak (-0.018, 0.247, 0.259) for the ATP dataset. The correlation between break points faced and aces (Figure Seven) is negative while the other two correlations are positive (Figure Eight & Nine). This predictor may be less stable than the others because of this high variability and may be a worse fit in the model. While double faults in the ATP and WTA show the smallest variability suggesting that most players commit around the same number of double faults.

Table One

Descriptive Statistics for Explanatory Variables for ATP

Variable	<i>n</i>	<i>Mean</i>	<i>SD</i>	<i>1st Q</i>	<i>Median</i>	<i>3rd Q</i>	<i>Skew</i>
Double Faults (x_1)	302	3.370	2.484	2	3	4	2.273
	6						

Break Points Faced (x_2).	647 9	7.215	4.287	4	6	10	.893
Aces (x_3)	651 6	7.256	5.242	3	6	10	1.511

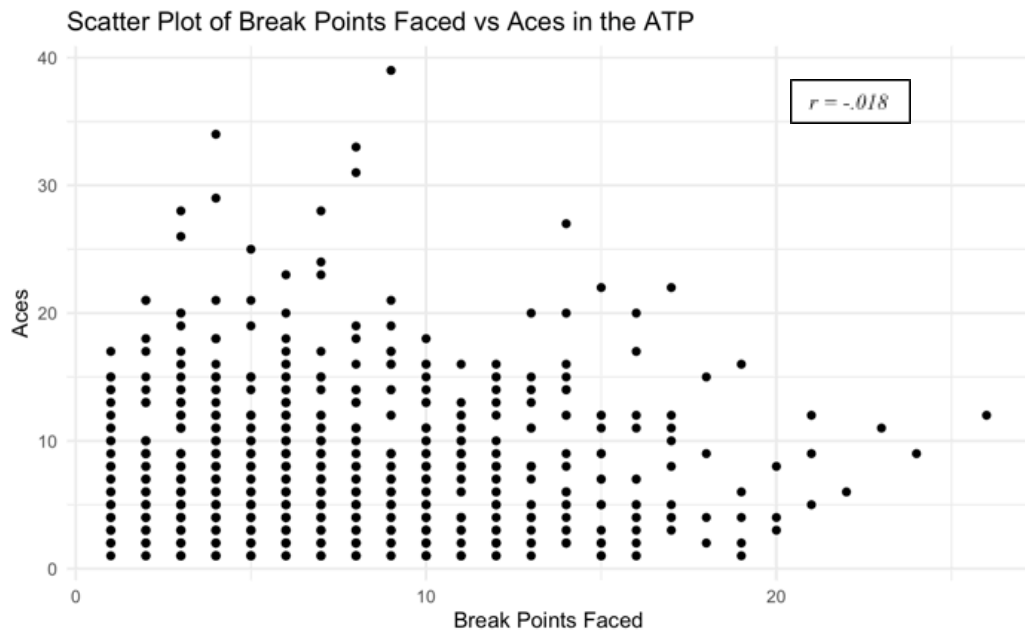
Table Two

Descriptive Statistics for Explanatory Variables for WTA

Variable	n	$Mean$	SD	$1^{st} Q$	Median	$3^{rd} Q$	Skew
Double Faults (x_1)	342 1	4.276	2.613	2	4	6	1.161
Break Points Faced (x_2).	650 1	8.126	4.079	5	8	11	.506
Aces (x_3)	348 1	4.351	3.192	2	4	6	1.464

Figure Seven

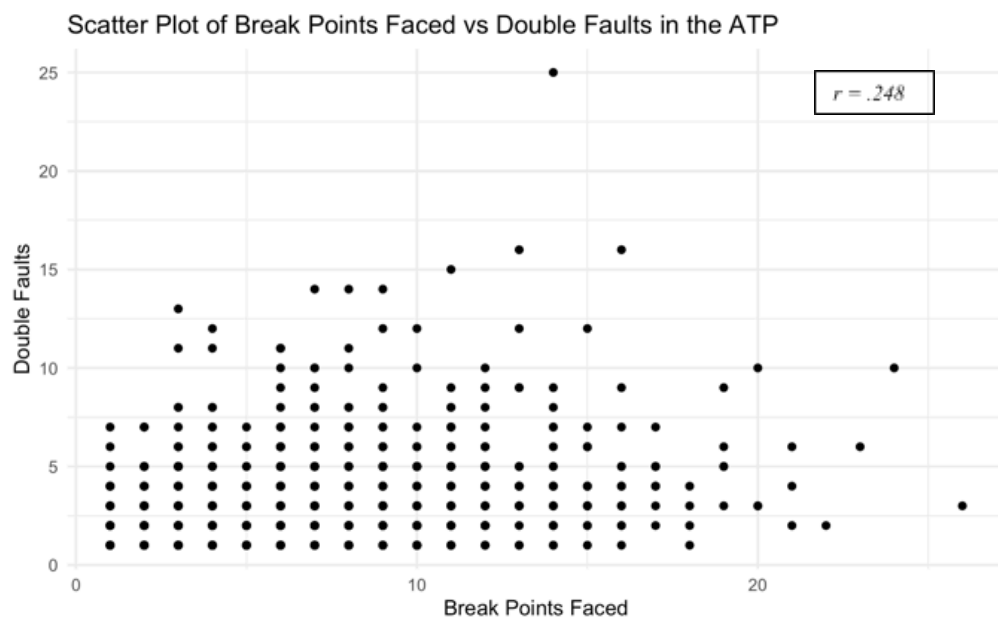
Scatterplot of Break Points Faced and Aces for ATP Players



Note. Scatterplot showing a weak negative correlation between break points faced and aces among ATP players ($r = -0.018$).

Figure Eight

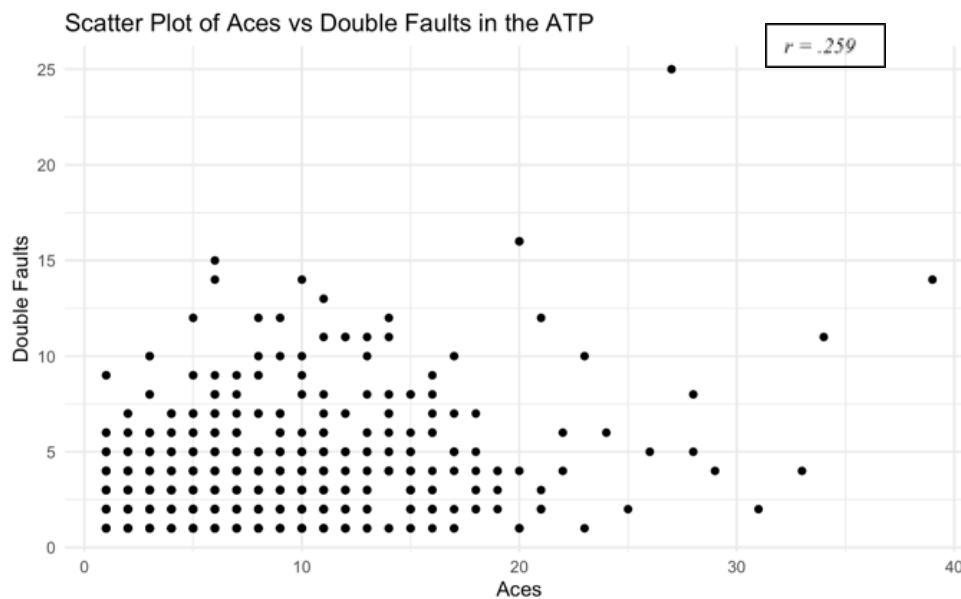
Scatterplot of Break Points Faced and Double Faults for ATP Players



Note. Scatterplot displaying a weak positive correlation between break points faced and double faults in ATP players ($r = 0.247$).

Figure Nine

Scatterplot of Aces and Double Faults for ATP Players

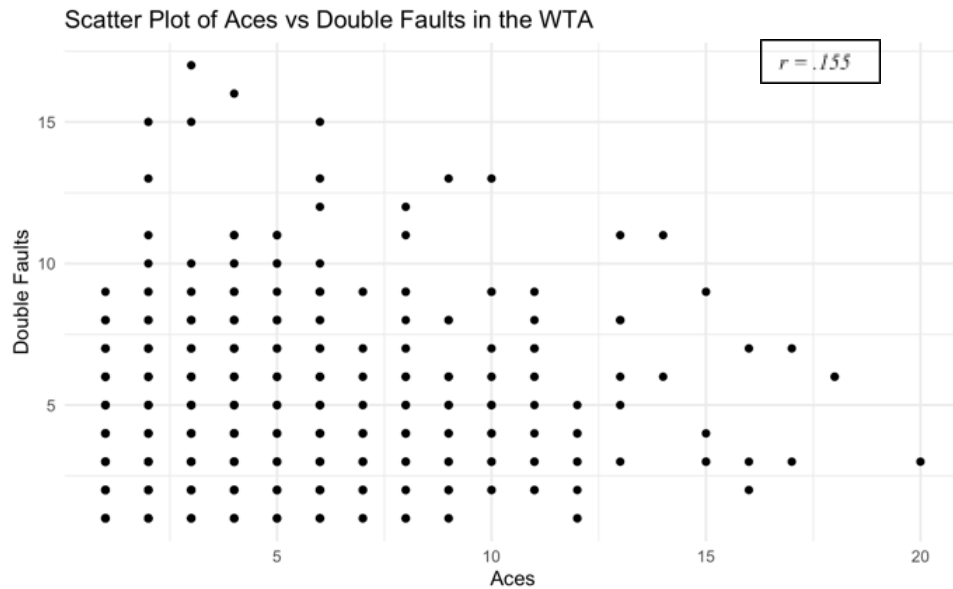


Note. Scatterplot illustrating a weak positive correlation between aces and double faults among ATP players ($r = 0.259$).

The WTA dataset showed weak and moderate correlations (-0.023, 0.420, 0.155). The correlation between aces and double faults was positive and weak (Figure Ten), the correlation between aces and break points faced was negative and weak (Figure Eleven) but the correlation between break points faced and double faults was a moderate positive correlation (Figure Twelve). The weak correlations in the ATP indicate that the number of break points faced, the number of double faults and the number of aces a player serves do not really depend on each other. The WTA on the other hand does experience some moderate correlations where the number of double faults a player has correlates with more break points faced. This is slightly worrying about some potential issues between break points faced and double faults as predictors in the regression model. Other than that every other correlation displayed that there may not be collinearity within the model. If all three predictors are significant in the model they would be explaining different parts of a match outcome.

Figure Ten

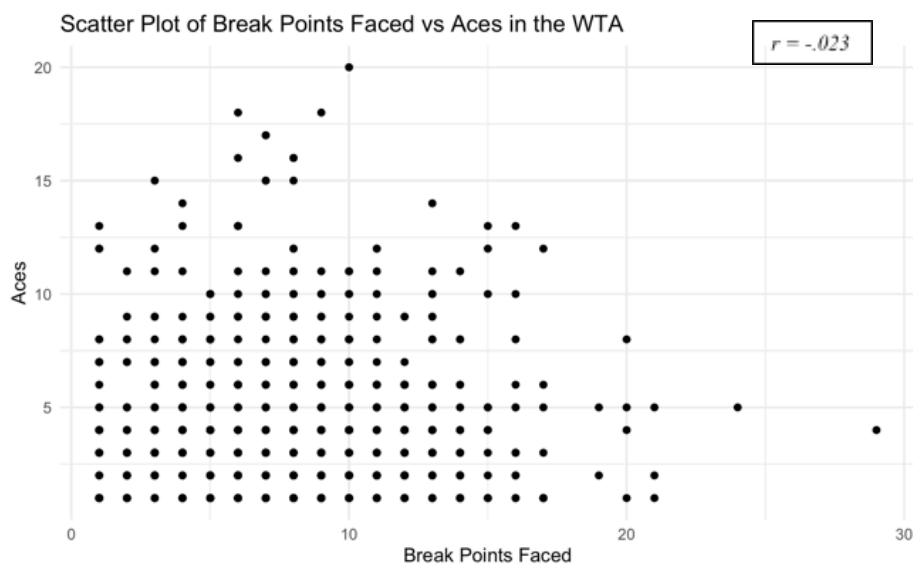
Scatterplot of Aces and Double Faults for WTA Players



Note. Scatterplot indicating a weak positive correlation between aces and double faults in WTA players ($r = 0.248$).

Figure Eleven

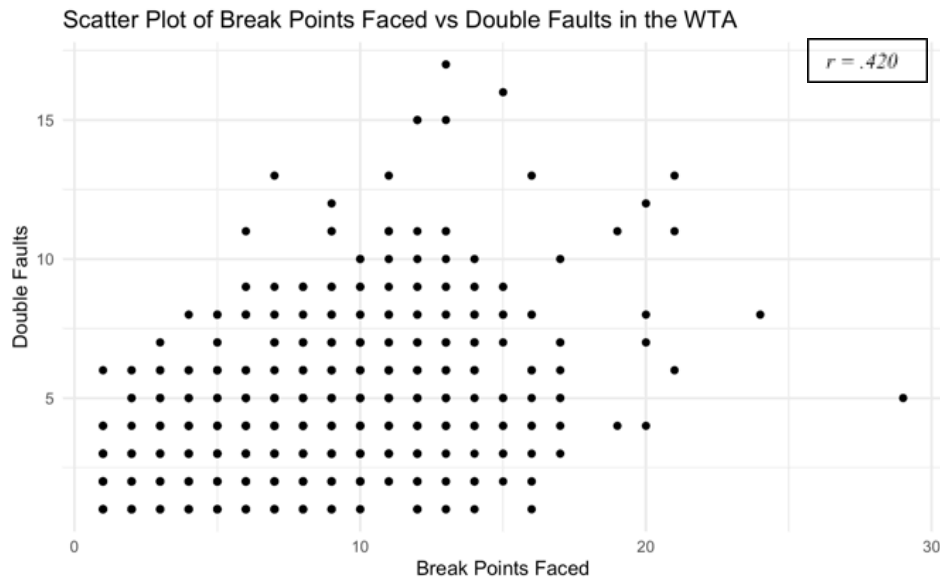
Scatterplot of Break Points Faced and Aces for WTA Players



Note. Scatterplot showing a weak negative correlation between break points faced and aces among WTA players ($r = -0.023$).

Figure Twelve

Scatterplot of Break Points Faced and Double Faults for WTA Players



Note. Scatterplot demonstrating a moderate positive correlation between break points faced and double faults in WTA players ($r = 0.420$).

Before diving deeper into the impact of each predictor the original logistic regression is $\log \log \left(\frac{P(win)}{(1-P(win))} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$. The VIF values for the both the ATP were ($x_1 = 1.129$, $x_2 = 1.045$, $x_3 = 1.084$) and the WTA values ($x_1 = 1.216$, $x_2 = 1.168$, $x_3 = 1.046$) were all not exceeding 4 therefore there was no issues with collinearity after considering the VIF values and the correlations. The moderate correlation between break points faced and double faults was noted but since there VIF values were all below 4 I decided to continue using both as a predictors in the WTA model. The correlations and VIF values indicate aces, double faults and break points faced are independent predictors of the model. Table Three and Four are the summaries of the logistic regressions.

One key difference between the WTA and ATP models was that double faults was insignificant in the logistic regression for the WTA (p-value = .0886) but significant for the ATP. To rectify this the WTA model was adapted to be

$$\log \log \left(\frac{P(\text{win})}{(1-P(\text{win}))} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \text{ where } x_1 \text{ is break points faced and } x_2 \text{ is}$$

aces. This reduced model also had a lower BIC value implying a better model fit then the full model for the WTA. A potential reason for the removal of double faults is its moderate correlation with number of break points faced which was originally ignored after the VIF values. Additionally, the coefficient for aces in the WTA is negative while it is positive for the ATP indicating that the chance of a loss still increases when an ATP player has more aces while in the WTA it decreases the chance of a loss. There are some potential reasons for the positive coefficient in the ATP one being that male tennis players commit a lot more serves and depend on aces much more than female players. While in the WTA aces are less important and normally only players who have great serves depend on aces as an element of their style of play. The ATP having quality serve is integral to be a top player.

Table Three

ATP Model Summary

Variable	Coefficien <i>t</i>	SE	<i>z</i>	<i>p</i>
Intercept	-1.030	0.186	-5.550	2.86×10^{-8}
Double Faults (x_1)	0.071	0.032	2.213	.0269
Break Points Faced (x_2).	0.180	0.020	9.051	$< 2 \times 10^{-16}$
Aces (x_3)	0.067	0.015	-4.434	9.27×10^{-6}

Table Four

WTA Model Summary (adapted)

Variable	<i>Coefficien</i>	<i>SE</i>	<i>z</i>	<i>p</i>
	<i>t</i>			
Intercept	-1.017	0.206	-4.932	8.13×10^{-7}
Break Points Faced (x_1)	0.183	0.209	8.759	$< 2 \times 10^{-16}$
Aces (x_2).	-0.107	0.035	-4.259	2.05×10^{-5}

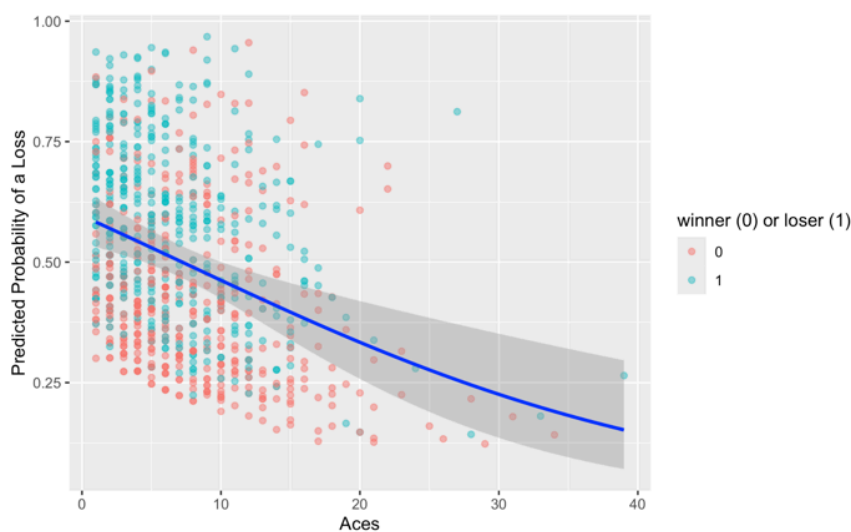
Five scatterplots were created to see the impact of each predictor and the predicted probability of a loss. The probability ranges from 0 (no chance of losing) to 1 (a certain loss) and was created from the models. Then, each observation is colour coded red or blue to represent actual match outcomes. Red is for match wins and blue for losses. This allows us to compare predicted probabilities with actual outcomes. There is a fitted logistic curve to every graph representing the predicted probability of a loss to each predictor. Lastly, there is a grey band around the logistic curve representing a 95% confidence interval.

The ATP and WTA showed similar logistic curves for all three predictors. The ATP model estimate for aces was positive (Table Three) implying that the more aces a player has the more likely they were to lose. However, in the graph between the predicted probability of a loss and aces there is a negative logistic curve implying the opposite. This observed trend reflects a more complex relationship between the variables in the dataset. Specifically, players who serve many aces may also exhibit other winning traits (fewer double faults or fewer breaks points faced) which could explain this negative curve (Figure Thirteen). Also, the scatterplot has an extreme skew demonstrating that regardless of match outcome most ATP players were hitting under 20 aces per match which could also be impacting the logistic curve.

For break points faced and double faults they both indicate a positive relationship which aligns with the model coefficients. Suggesting that the more break points faced or double faults a player has in the ATP the more likely they are to lose. The graph with break points faced shows the best logistic curve alignment reflecting how impactful it is on the model. Additionally, at around 5 break points faced in the ATP the points become majority blue implying losses (Figure Fourteen). The graph for double faults demonstrates that it was a weak predictor for the probability of a loss. There is a outlier value where one player committed 25 double faults potentially skewing the logistic curve and the model (Figure Fifteen).

Figure Thirteen

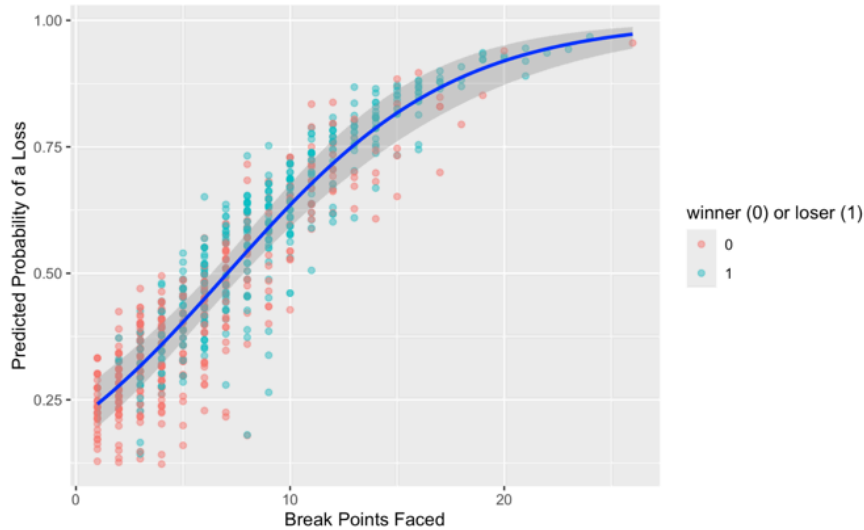
Logistic Curve for Predicted Probability of Loss Based on Aces for ATP Players



Note. Logistic regression curve with a 95% confidence band, showing predicted loss probability as a function of aces for ATP players.

Figure Fourteen

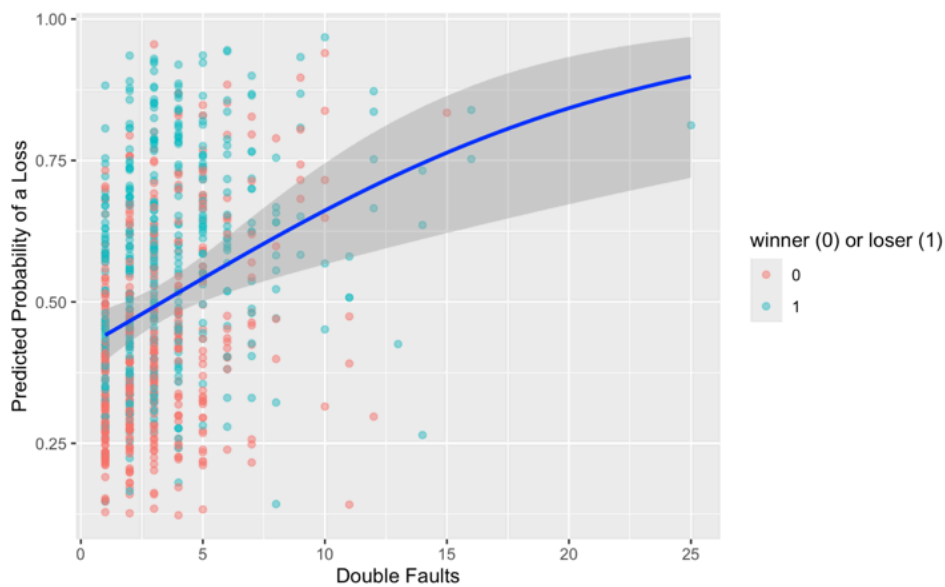
Logistic Curve for Predicted Probability of Loss Based on Break Points Faced for ATP Players



Note. Logistic regression curve with a 95% confidence band, indicating a positive relationship between break points faced and loss probability for ATP players.

Figure Fifteen

Logistic Curve for Predicted Probability of Loss Based on Double Faults for ATP Players



Note. Logistic regression curve with a 95% confidence band, showing a positive but weak relationship between double faults and loss probability for ATP players.

The WTA dataset reveals logistic curves for aces and break points faced that are similar to those observed in the ATP dataset. However, differences in predictor significance slightly impact the interpretation. For the WTA, aces demonstrate a negative relationship in

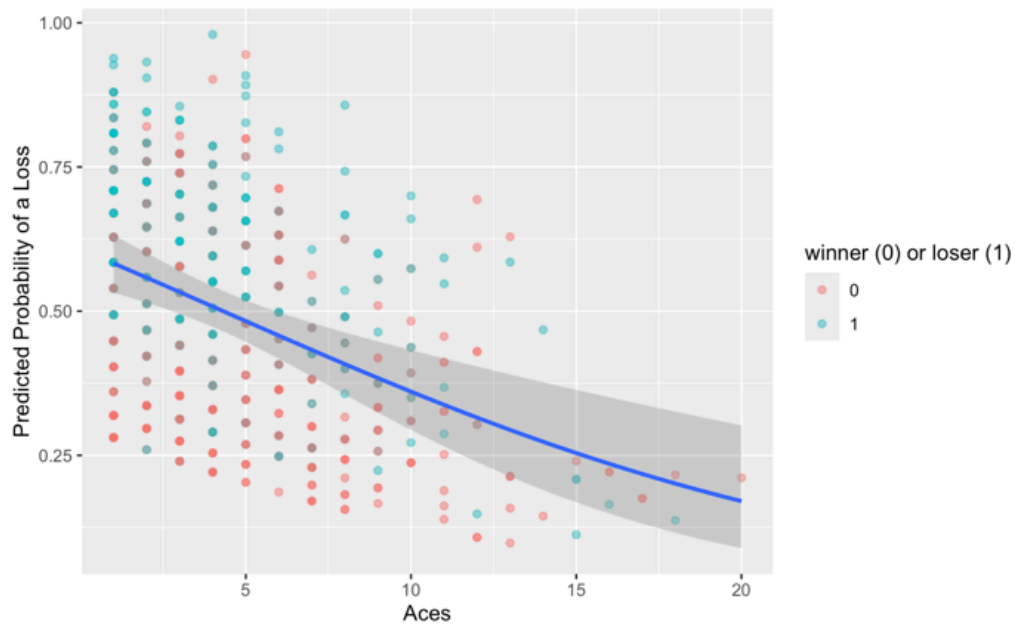
both the model coefficient and the logistic curve, implying that the more aces a player hits, the less likely they are to lose (Figure Sixteen). This trend is more consistent than in the ATP, where the relationship was counterintuitive. This alignment between the coefficient and curve suggests that in the WTA dataset, aces may better encapsulate player performance without as many confounding factors as in the ATP.

The logistic curve for break points faced aligns well with the WTA model's coefficient, indicating a positive relationship (Figure Seventeen). This suggests that the more break points a WTA player faces, the higher the probability of losing a match. As seen in the graph, the logistic curve steepens significantly, and by around 5-10 break points faced, there is already a shift toward the probability of a loss dominating, marked by more blue points representing losses similar to the ATP dataset. This suggests that break points faced is the strongest predictor of match outcome in the WTA dataset as well.

The absence of double faults as a significant predictor simplifies the WTA model compared to the ATP. This difference might reflect the higher variability in other factors contributing to match outcomes in the ATP.

Figure Sixteen

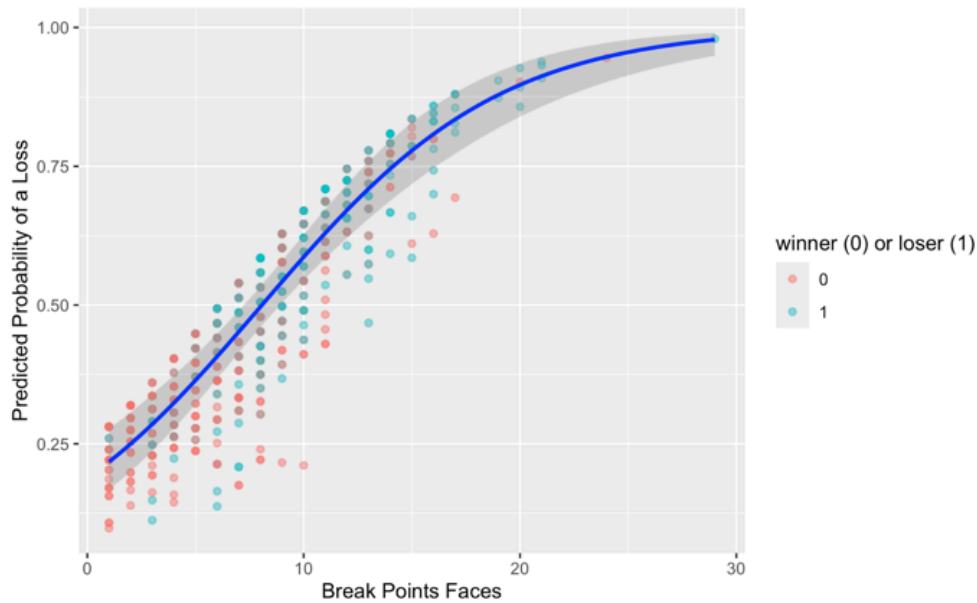
Logistic Curve for Predicted Probability of Loss Based on Aces for WTA Players



Note. Logistic regression curve with a 95% confidence band, highlighting a negative relationship between aces and loss probability for WTA players.

Figure Seventeen

Logistic Curve for Predicted Probability of Loss Based on Break Points Faced for WTA Players



Note. Logistic regression curve with a 95% confidence band, illustrating a strong positive relationship between break points faced and loss probability for WTA players.

To assess the model accuracy in predicting a match outcome three confusion matrices were created. One for the ATP, another for the WTA and lastly a merged dataset to improve the statistical power and generalizability of the model. The ATP model is more accurate and precise, but the WTA model has a higher recall percentage and F1 score. Additionally, a McNemar's Test was conducted to see if the model's misclassifications are balanced. The ATP model rejected the null ($p\text{-value} = 0.017$) implying that there is bias in the model. The WTA failed to reject the null ($p\text{-value} = 0.95$) implying a more balanced model. Both models tend to have bias in predicting more winners than losers. Merging the datasets took the best qualities of both models. The merged matrix was more accurate and precise than the WTA model while also having a higher recall percentage and F1 score than the ATP. Lastly, the merged matrix failed to reject the null ($p\text{-value} = .07$) like the WTA model suggesting it is a balanced model.

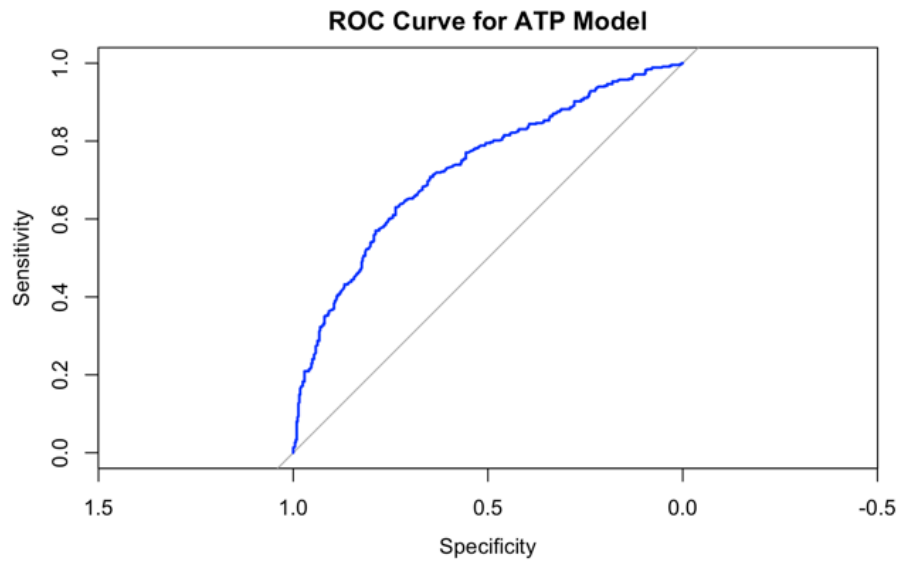
The ROC (Receiver Operating Characteristic) curves for the ATP and WTA models help evaluate how well the logistic regression models predict match outcomes. These curves show the relationship between sensitivity (true positive rates) and 1-specificity (false positive

rates) at different thresholds (different likelihoods for a winning and losing a match). The gray diagonal line represents a model that makes random guesses. A good model will have a curve that quickly rises toward the top-left corner, indicating high accuracy with few false positives. The area under the curve (AUC) is a critical metric derived from these graphs, measuring the overall model performance. Higher AUC values for the models suggest stronger predictive capabilities.

In the ATP model, the ROC curve exhibits a consistent upward trend, deviating significantly from the diagonal line, indicating good model performance (Figure Eighteen). Similarly, the WTA model's ROC curve shows similar behaviour, though it appears less steep in parts, suggesting that the WTA model may be slightly less effective at distinguishing winners and losers compared to the ATP mode (Figure Nineteen). This could stem from differences in data structure or the predictive power of features across datasets. The areas under the curve for both models was in the good range (.7283 for the ATP and .7137 for the WTA) which suggests that these are solid models at predicting match outcomes. However, there is room for improvement.

Figure Eighteen

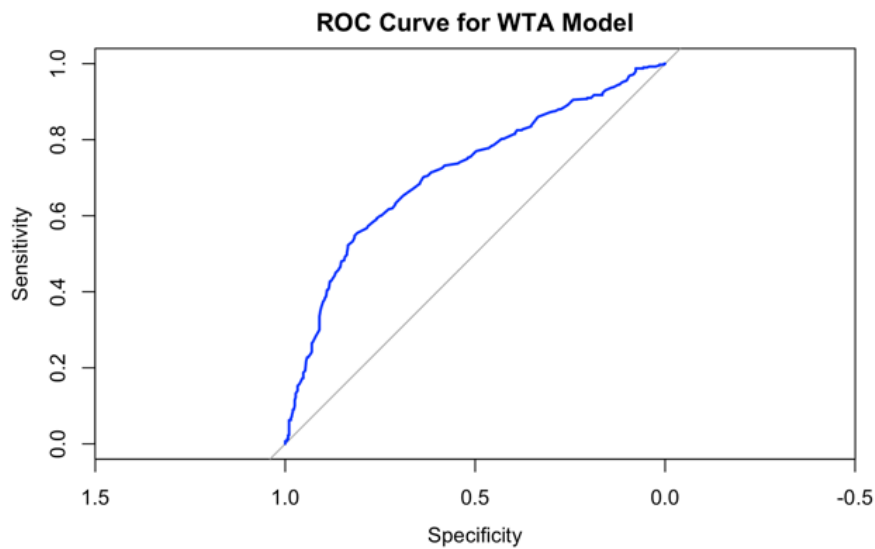
ROC Curve for ATP Logistic Regression Model



Note. Receiver Operating Characteristic (ROC) curve for the ATP model, with an area under the curve (AUC) of 0.7283, indicating good predictive performance.

Figure Nineteen

ROC Curve for WTA Logistic Regression Model



Note. ROC curve for the WTA model, with an AUC of 0.7137, reflecting solid predictive capabilities with room for improvement.

Conclusion

This study examined the impact of three key performance metrics: double faults, break points faced, and aces on match outcomes for ATP and WTA players, focusing specifically on the top 50 ranked players in professional tennis. Logistic regression models

were used to identify patterns in match outcomes which led to the discovery that there are gendered differences to what influences match outcome. Break points faced consistently regardless of gender was the most significant predictor for match outcomes. Double faults were an insignificant metric in the WTA but significant in the ATP. Aces had contrasting results for match outcomes in the ATP and the WTA. In the ATP more aces meant more of a likelihood of a loss while in the WTA more aces meant more of a likelihood of a win. These differences and similarities highlight the distinct styles of play and dynamics between two of the tours.

The practical implications of these findings can lead to performance optimization and gender-specific strategies. Coaches should coach players to avoid facing break points and getting broken. For female players encouraging a strong consistent ace could be productive while for male players less aces and focusing more on other parts of the game could be beneficial. Predictive models developed in this study could serve as foundational tools for real-time decision-making during matches. This study had several limitations from restricted scope and data exclusions. By focusing only on the top 50 players, the findings may not generalize to lower-ranked players or doubles matches. Matches with zero values for the predictors were omitted, potentially introducing selection bias. Future research could include data from lower ranked players and consider the service game's from doubles matches (mixed doubles and classic doubles). Additionally, including factors such as serve speed, unforced errors and opponent rankings could make a more powerful predictive model. Analyzing matches depending on surface may also reveal surface differences to a successful service game.

This study highlights the nuanced differences and similarities between the ATP and WTA at the highest level while identifying pathways for players to sway matches in their

favour. These insights can be expanded to a broader spectrum of tennis athletes and match scenarios through further research and addressing the limitations,

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